American Computer Science League

2020-2021 ● Contest 2: Shorts ● Senior Division

1. Prefix-Infix-Postfix

Convert this expression from prefix to postfix:

− + A ^ B 2 ^ / * A C + B C 3

A. AB2^+A*CBC+/3^−
B. AB2^+AC*BC/+3^−
C. AB2+^AC*BC+/3^−
D. AB2^+AC*BC+/3^−
E. AB2^+AC*BC+/3−^

2. Prefix-Infix-Postfix

What is the value of the following prefix expression if x = -5 and y = 2?

− * 4 ^ x y − * 9 x ^ 6 y

A. -181
B. -29
C. 29
D. 157
E. 181

3. Bit-String Flicking

Evaluate the following:

((RCIRC-3 01110) XOR 10111 AND 01111 OR (LSHIFT-2 11101))

A. 01110
B. 11110
C. 01011
D. 11011
E. 10100

4. Bit-String Flicking

Identify the actual 5-character bit string that makes the following expression TRUE using * notation (e.g 0*10*):

(RSHIFT-1 ((RCIRC-2 01101) AND X)) = (LCIRC-4 00010)

A. *1*1*
B. *0*0*
C. *0*1*
D. ***1*
E. *0***

5. LISP

Evaluate the following LISP expression:

(CDR (CAR (CDR (CDR '(a (b (c d)) (e f g))))))

A. (e)
B. (g)
C. (e f)
D. (f g)
E. (e g)